

ATP Creativity I

Isolating and Controlling Creativity in Automated Theorem Proving

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Abstract

This paper proposes an experimental framework for isolating and controlling creativity in automated theorem proving. Creativity is operationalized as the production of a verified proof that is structurally novel under fixed information, training, formal-environment, and computational-resource constraints. The framework combines controlled proof-search parameters with the proof-counting functions and proof-space upper bound introduced in Section 8 of *Depths of Induction and Formalized Self-Awareness*. The result is a target-relative measure of creative yield that distinguishes verified discovery from randomness, memorization, brute-force enumeration, and merely increased computation.

1 Operationalizing Creativity

Creativity is not treated here as an unanalyzed metaphysical faculty. Instead, the experimentally measurable substitute is:

Creativity is the production of a verified proof that is structurally novel under controlled information and equal resource constraints.

For an automated theorem prover, the essential division is

The verifier determines correctness; the search policy supplies creativity.

The verifier should remain deterministic and noncreative. It accepts or rejects proof certificates. Creativity is located entirely in the prover's selection and organization of legal proof-search actions.

2 Freezing the Experimental Environment

To isolate creativity, all factors other than search behavior must be held fixed. Across compared runs, freeze:

1. the formal system $F = (x, y, z)$;
2. the hypothesis set $\Gamma \subseteq y$;
3. the target theorem $\varphi \in y$;
4. the available axioms, definitions, and theorems;
5. the legal proof-state and action spaces;

6. the proof verifier;
7. the model architecture and weights;
8. the training data and preprocessing procedure;
9. the time, memory, inference, and node budgets;
10. the proof-acceptance and normalization criteria.

Without these controls, an apparent increase in creativity may instead be caused by additional knowledge, greater compute, a larger theorem library, or a weaker verification standard.

3 Information Conditions

Let the experiment include three information conditions.

Condition	Information supplied	Interpretation
K_0	Target and frozen formal environment only	Independent proof discovery
K_1	Broad theorem families or decomposition hints	Guided structural discovery
K_2	Information correlated with a known proof route	Guided reconstruction or completion

In condition K_0 , the ATP must receive no target-specific score based on distance from the reference proof graph. A proof discovered in K_2 may still be novel, but it should not be described without qualification as an independent rediscovery.

4 The Creativity-Control Vector

Let the search policy be governed by a control vector

$$\mathbf{c} = (\tau, w, b, n, r, \lambda, \ell),$$

where:

τ	action-selection temperature;
w	candidate width;
b	breadth-versus-depth allocation;
n	novelty bonus;
r	restart or branch-abandonment rate;
λ	willingness to construct intermediate lemmas;
ℓ	proof-length penalty.

Each component should first be varied independently. After calibration, define a single master creativity parameter

$$c \in [0, 1], \quad c \mapsto \mathbf{c}(c).$$

Typical interpretations are:

- $c = 0$: nearly deterministic exploitation,
- $c = \frac{1}{2}$: balanced exploitation and exploration,
- $c = 1$: maximal structured exploration within the fixed budget.

Increasing c must not increase the total computational budget. It may only redistribute a fixed amount of search effort.

5 Creativity Is Not Randomness

A search policy that enters unusual proof states but produces no proof is merely noisy. A successful run should therefore receive credit only when it produces a verified certificate.

For a proof P , define the preliminary creativity indicator

$$\text{CI}(P) = \mathbf{1}_{\text{verified}(P)} \cdot \text{Independence}(P) \cdot \text{StructuralNovelty}(P).$$

The structural-novelty component may incorporate:

- a. normalized proof-DAG distance from a reference proof;
- b. the proportion of predecessor theorems absent from the reference route;
- c. use of a different cardinality or algebraic decomposition;
- d. creation of reusable intermediate lemmas;
- e. use of a different explicit injection, surjection, or bijection;
- f. bypassing expected lemmas by unrelated formal machinery;
- g. diversity among verified proofs obtained from different random seeds.

Before novelty is measured, proofs should be normalized by removing redundant detours, duplicate steps, syntactic padding, and irrelevant permutations.

6 Section 8 Proof Counting

Fix a formal system $F = (x, y, z)$, hypotheses $\Gamma \subseteq y$, and a target $\varphi \in y$.

For $n \in \mathbb{N}^+$, define

$$P_F(\Gamma, \varphi, n)$$

to be the number of proofs of φ from Γ having length exactly n .

Let

$$\|\varphi\|_F(\Gamma) := \min\{|\pi| : \pi \text{ is a proof from } \Gamma \text{ with final line } \varphi\}$$

when $\Gamma \vdash_F \varphi$, and let $\|\varphi\|_F(\Gamma) = \infty$ otherwise.
The number of shortest proofs is

$$P_F^{\min}(\Gamma, \varphi) := P_F(\Gamma, \varphi, \|\varphi\|_F(\Gamma))$$

when $\|\varphi\|_F(\Gamma) < \infty$, and

$$P_F^{\min}(\Gamma, \varphi) := 0$$

otherwise.

Assume that the maximum rule arity in F is $a \in \mathbb{N}^+$. For each $k \in \{1, \dots, a\}$, let

$$\kappa_k := |z_k(F)|$$

be the number of k -ary inference rules.

Define the Section 8 proof-space envelope

$$B_F(\Gamma, n) := \prod_{j=1}^n \left(|\Gamma| + \sum_{k=1}^a \kappa_k (j-1)^k \right).$$

Then

$$\boxed{P_F(\Gamma, \varphi, n) \leq B_F(\Gamma, n)}$$

and, when

$$m = \|\varphi\|_F(\Gamma) < \infty,$$

we obtain

$$\boxed{P_F^{\min}(\Gamma, \varphi) \leq B_F(\Gamma, m)}.$$

The bound is intentionally crude. It counts many line sequences that are not valid proofs and many construction plans that collapse to the same formulas. Its value is that it supplies a common combinatorial envelope derived from the hypothesis cardinality, rule cardinalities, rule arities, and proof length.

7 Shortest-Proof Scarcity

For a verified theorem φ with

$$m = \|\varphi\|_F(\Gamma) < \infty,$$

define the shortest-proof scarcity measure

$$\boxed{R_F(\Gamma, \varphi) := \log_2 \left(\frac{B_F(\Gamma, m)}{P_F^{\min}(\Gamma, \varphi)} \right)}.$$

Equivalently,

$$R_F(\Gamma, \varphi) = \log_2 B_F(\Gamma, m) - \log_2 P_F^{\min}(\Gamma, \varphi).$$

A theorem with many shortest proofs has lower shortest-proof scarcity. A theorem with only one or a few shortest certificates has greater shortest-proof scarcity relative to the same Section 8 proof-space envelope.

This quantity is not an exact entropy and does not presuppose that the ATP samples proofs uniformly. It is a target-relative combinatorial index.

8 Controlled Discovery and Novelty

Let

$$\pi_c$$

be the ATP search policy at creativity setting $c \in [0, 1]$.

Let:

p = the controlled fraction or level of general proof-training exposure,

K_j = the information condition,

and

T = the fixed resource budget.

Define the verified discovery probability

$$D(c; p, K_j, T) := \Pr_{\pi_c}(\text{a verified proof of } \varphi \text{ is found within } T).$$

Assign each normalized verified proof P a structural-novelty score

$$\nu(P) \in [0, 1].$$

Define conditional expected novelty by

$$N(c; p, K_j, T) := \mathbb{E}_{\pi_c}[\nu(P) \mid \text{the run succeeds}].$$

The unweighted creative yield is

$$\boxed{Y(c; p, K_j, T) = D(c; p, K_j, T)N(c; p, K_j, T).}$$

This product gives approximately zero credit in either of two cases:

$$D \approx 0 \implies \text{unproductive randomness,}$$

and

$$N \approx 0 \implies \text{routine reconstruction of a familiar proof.}$$

Verified and structurally novel proof discovery requires both factors to be positive.

9 Section 8–Weighted Creative Yield

The proof-counting framework can be combined with the creativity controls by defining

$$Y_F^*(c; p, K_j, T) = D(c; p, K_j, T)N(c; p, K_j, T)R_F(\Gamma, \varphi).$$

Expanding the scarcity term gives

$$Y_F^*(c; p, K_j, T) = D(c; p, K_j, T)N(c; p, K_j, T) \log_2 \left[\frac{\prod_{j=1}^m \left(|\Gamma| + \sum_{k=1}^a \kappa_k (j-1)^k \right)}{P_F^{\min}(\Gamma, \varphi)} \right],$$

where

$$m = \|\varphi\|_F(\Gamma).$$

The three factors have distinct roles:

$$\underbrace{D}_{\text{successful discovery}} \quad \underbrace{N}_{\text{structural novelty}} \quad \underbrace{R_F}_{\text{target scarcity}}.$$

The first two are properties of the prover under the experimental condition. The third is a property of the benchmark target relative to the formal system and the Section 8 envelope.

10 The Isolation Experiment

For a fixed benchmark, hold

$$(F, \Gamma, \varphi, p, K_j, T)$$

constant and vary only the master creativity parameter c .

Define the isolated creativity effect relative to deterministic exploitation by

$$\Delta Y_F^*(c) = Y_F^*(c; p, K_j, T) - Y_F^*(0; p, K_j, T).$$

Because $R_F(\Gamma, \varphi)$ is constant across the compared runs,

$$\Delta Y_F^*(c) = R_F(\Gamma, \varphi) [D(c; p, K_j, T)N(c; p, K_j, T) - D(0; p, K_j, T)N(0; p, K_j, T)].$$

Thus, under the frozen protocol, a difference in creative yield cannot be attributed to a larger theorem library, increased computation, additional route information, or a different formal problem. It is attributable to the controlled change in search behavior.

11 Repeated Matched Trials

For every experimental cell

$$(p, c, K_j),$$

perform many runs using prespecified seeds. Record at least

$$(\text{outcome}, \text{nodes}, \text{time}, \text{proof DAG}, \text{novel lemmas}, \text{seed}).$$

The outcome categories should be:

1. verified proof;
2. verified negation;
3. honest exhaustion of the fixed budget;
4. protocol or implementation failure.

The principal estimands are

$$D(c; p, K_j, T), \quad N(c; p, K_j, T), \quad Y(c; p, K_j, T), \quad Y_F^*(c; p, K_j, T).$$

Confidence intervals should be estimated across independent seeds. Proof novelty should be evaluated only after verification and normalization.

12 Expected Creativity Curve

A plausible empirical pattern is an inverted-U relationship:

$$\text{very low } c \implies \text{overexploitation of familiar routes,}$$

$$\text{moderate } c \implies \text{productive exploration and reusable lemma discovery,}$$

$$\text{very high } c \implies \text{poorly directed branching and reduced success.}$$

The main empirical objective is therefore to estimate

$$\boxed{c^* \in \arg \max_{c \in [0,1]} Y_F^*(c; p, K_j, T).}$$

For a fixed target, $R_F(\Gamma, \varphi)$ is constant, so this is equivalent to maximizing

$$D(c; p, K_j, T)N(c; p, K_j, T).$$

Across different targets, the scarcity term permits a principled target-relative weighting.

13 The Password Analogy and Its Limit

A proof resembles a password candidate in one important respect: once a candidate proof is exhibited, local verification may be much easier than search.

The analogy is limited because formal proofs possess reusable internal structure. Intermediate lemmas, partial derivations, macro rules, theorem graphs, search order, and retained subproofs can improve later search. A wrong partial password usually has no comparable value.

Controlled ATP creativity is therefore not random guessing. It is the ability to allocate a fixed search budget so that reusable formal structure is discovered and exploited in the production of rare, valid, and unfamiliar certificates.

14 Principal Research Question

The framework reduces the central experimental question to:

At what controlled exploration level does verified structural novelty emerge most efficiently relative to the Section

This formulation isolates three logically separate issues:

1. whether the ATP succeeds;
2. whether the successful proof is structurally novel;
3. how scarce the target's shortest proofs are relative to a common combinatorial proof-space bound.

Accordingly, the framework distinguishes creativity from correctness, randomness, memorization, brute-force enumeration, and additional computation.